

IIT Mandi iHub and HCI Foundation*

Technology in Harmony with Human Needs

Multimodal AI Hackathon



*Part of the Technology Innovation Hub (TIH)
National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS)

HAPPY SUNNY DAY

Unimodal depiction



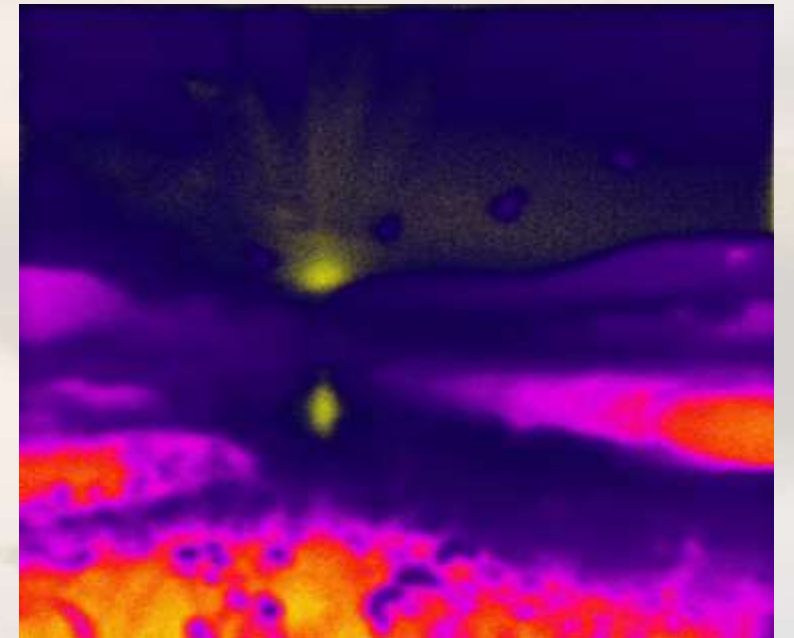
Image



Video



Audio



Thermal

Multimodal Depiction

What is Multimodal AI ?

- A **paradigm shift** in artificial intelligence.
- Moving beyond single-data-type processing to **integrate and synthesise diverse information** streams simultaneously.
- A sophisticated way mirroring how **humans naturally perceive** and interpret the world around us.



Text

Natural language processing
and understanding



Images

Computer vision and visual
recognition



Audio

Speech and sound
analysis



Video

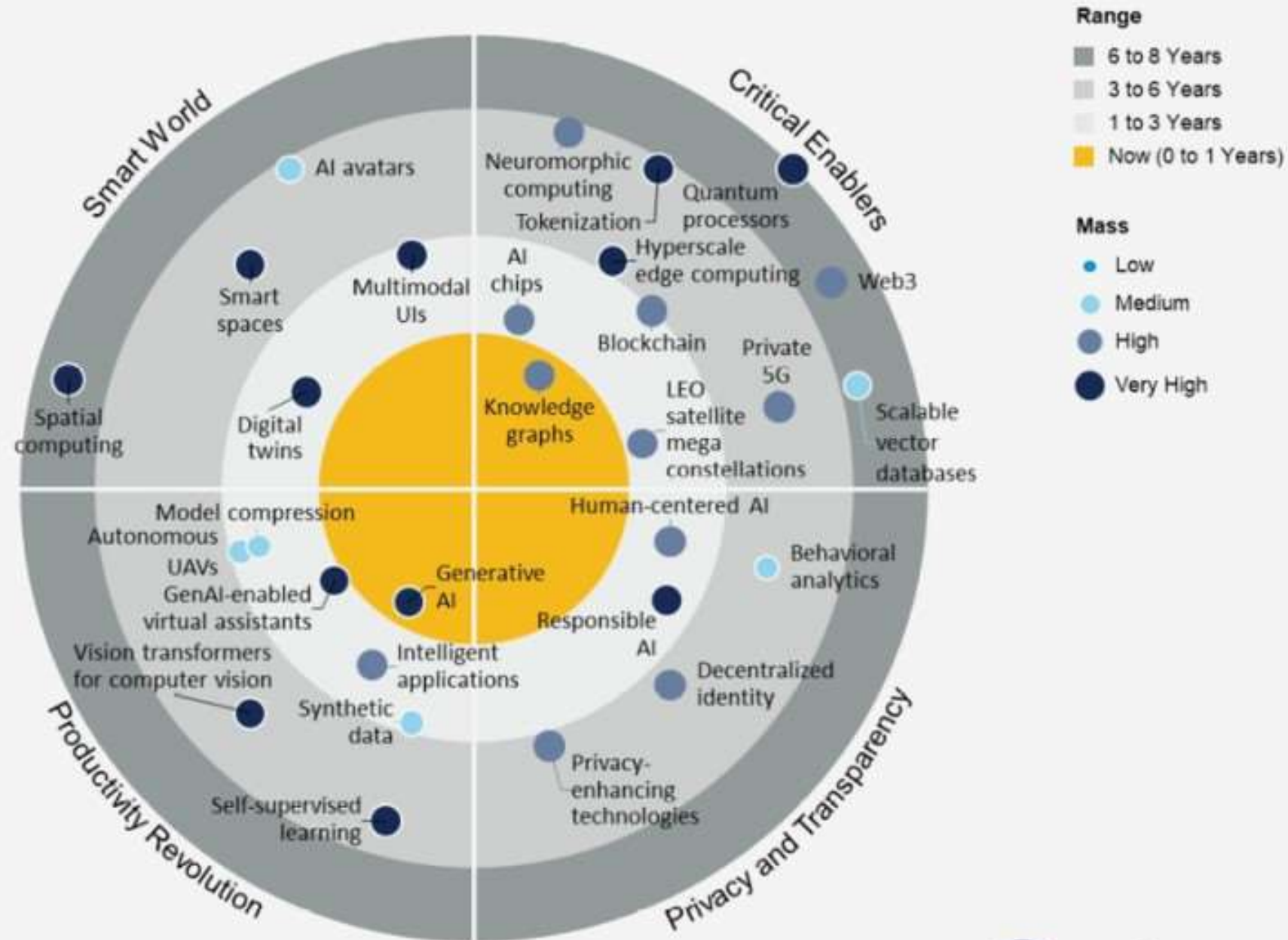
Motion and
temporal
understanding



Sensors

Environmental &
Physical data

Impact Radar for 2024

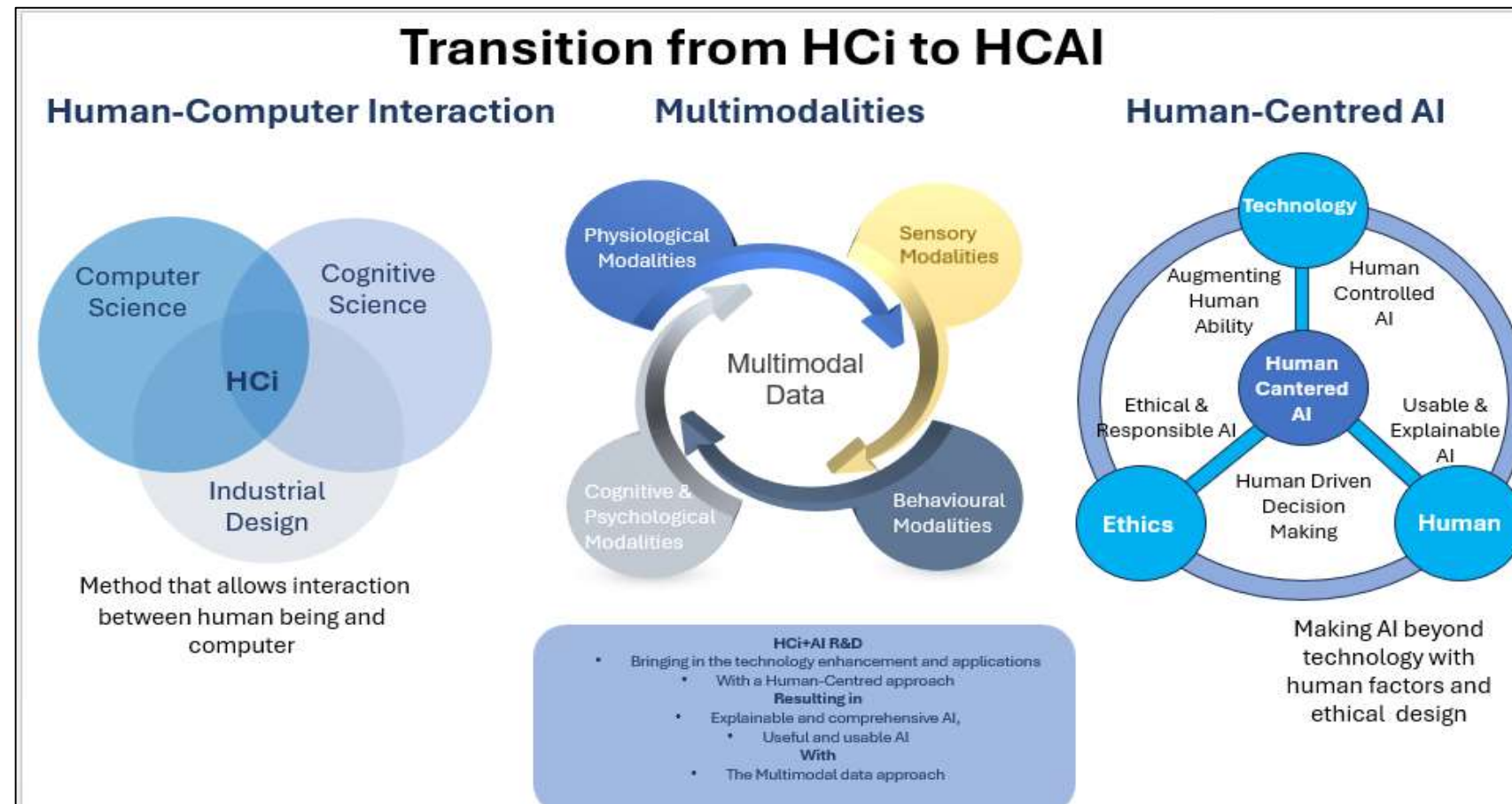


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Gartner

By processing multiple data types concurrently, Multimodal AI

- Achieve **human-like reasoning and perception**
- These systems don't just see or hear - they **comprehend context by synthesizing information** from multiple senses working in harmony
- Much like the **human brain processes** sight, sound, touch, and language simultaneously to form complete understanding.



Why Multimodal AI Matters

Real world challenges are inherently multimodal

These demand **understanding and integrating multiple information** streams **simultaneously** for effective problem-solving

Traditional AI systems operate in isolation, processing **single data** types **without contextual** awareness.

Real-World Applications Require Multimodality



Healthcare Diagnosis

Combines medical imaging (X-rays, MRIs), electronic patient records, doctor's audio notes, and real-time vital signs from sensors to provide comprehensive diagnostic insights that no single modality could achieve alone.



Autonomous Vehicles

Processes visual data from cameras, audio cues (sirens, horns), LIDAR sensor readings, GPS data, and traffic information simultaneously to navigate safely through complex, dynamic environments.



Smart Assistants

Understands voice commands whilst interpreting visual context from cameras, processing user location data, and analysing behavioural patterns to deliver contextually relevant, personalised responses.

The Hackathon Purpose

This 14 days long **collaborative event** is designed to **foster innovation in multimodal AI** whilst building a vibrant community of researchers, developers, and creative problem-solvers at IIT Mandi.

Encourage Exploration



Dive deep into cutting-edge multimodal AI research, experimenting with novel combinations of data types and emerging techniques that push the boundaries of what's possible.

Identify Talent



Discover exceptional participants who demonstrate innovative thinking and technical prowess, offering pathways to internships and research opportunities at iHub.

Human-Centred Platform



Provides a collaborative space where advanced technology meets thoughtful, user-focused design principles to create solutions that genuinely improve lives.

Strengthen Ecosystem



Build and reinforce IIT Mandi's innovation ecosystem by connecting students, faculty, industry experts, and researchers in meaningful collaborative relationships.

What You'll Build

Participants must select one of two distinct development paths, each offering unique opportunities to demonstrate technical skill and creative problem-solving in multimodal AI

Path 1: Indian Context Implementation

Implement a proof-of-concept that adapts existing **multimodal AI** technology specifically for **Indian scenarios and challenges**. Consider linguistic diversity, infrastructure constraints, cultural contexts, and unique local needs.

Examples: Voice assistants supporting regional languages, agricultural tech for smallholder farmers, healthcare solutions for rural areas, or traffic management for Indian road conditions.

Path 2: Novel Application

Propose and build an entirely **new application that leverages multimodal AI capabilities**—combining audio, video, generative AI, sensor data, or other modalities in innovative ways never seen before.

Examples: New forms of accessibility tools, creative expression platforms, educational experiences, environmental monitoring systems, or social connection applications.

Problem Statement Guidelines

A strong problem statement forms the foundation of your hackathon submission. Your proposal should comprehensively address these critical dimensions to demonstrate both technical depth and practical viability.

01

Relevance

How does your solution address a genuine, pressing real-world problem? Who experiences this problem, and what are the current pain points or limitations of existing approaches?

02

Multimodal Integration

Which specific modalities (text, image, audio, video, sensors) are you combining? Why is each modality essential? How do they complement each other to create understanding impossible with single modalities?

03

Differentiation

What distinguishes your approach from existing solutions? Are you applying known techniques to new contexts, combining modalities in novel ways, or introducing entirely new methodologies?

04

Feasibility

Can you realistically build a functional, demonstrable prototype within the 10-day development window? What are your technical requirements, and do you have access to necessary resources, data, and expertise?

05

Impact

Who are the primary beneficiaries of your solution? What measurable improvements will they experience? Consider scale, accessibility, cost-effectiveness, and long-term sustainability of your proposed impact.

Representative Sample

1. Inspiration: “EmoSense” Companion

The Challenge

Mental health challenges often develop gradually, with early warning signs going unnoticed until conditions become severe. Traditional monitoring relies on infrequent check-ins or self-reporting, which can be unreliable or delayed.

Who Benefits

Individuals managing stress, anxiety, or depression; caregivers monitoring loved ones; mental health professionals seeking continuous patient insights.

The Multimodal Solution

Computer Vision



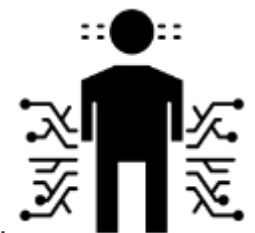
Analyses facial expressions, micro-expressions, and body language to detect emotional states and changes in baseline mood patterns over time.

Audio Processing



Evaluates voice tone, pitch, speaking rate, and linguistic patterns that correlate with emotional states and mental health indicators.

Physiological Sensors



Monitors heart rate variability, skin conductance, and other biometric markers that indicate stress, anxiety, or emotional dysregulation.

Outcome: EmoSense provides real-time emotional support with personalised interventions delivered through text recommendations, calming voice guidance, or visual mindfulness cues—all adapted to the user's current emotional state detected across multiple modalities.

Representative Sample

2.Inspiration: Culinary Vision AI

The Problem

Home cooking requires mastering techniques that are difficult to learn from static recipes or videos alone. Timing, temperature, and technique are nuanced skills traditionally passed down through hands-on teaching—a luxury many modern learners lack.

Computer Vision



Analyses cooking technique, ingredient preparation, browning levels, and visual doneness cues in real-time.

Audio Analysis



Listens to sizzling sounds, boiling intensity, and pan acoustics to determine optimal cooking temperatures and doneness.

Impact: An intelligent cooking companion that guides users through recipes with multi-sensory precision, making expert-level cooking techniques accessible to beginners whilst helping experienced cooks perfect their craft through objective, real-time feedback.

The Innovation

CulinaryVision transforms cooking into an interactive, guided experience by providing real-time, multi-sensory feedback that adapts to each cook's actions and the food's actual state rather than rigid recipe timings.

Smell Sensors



Detects aromatic compounds to suggest seasoning adjustments and alert to burning before visual signs appear.

Multimodal Feedback

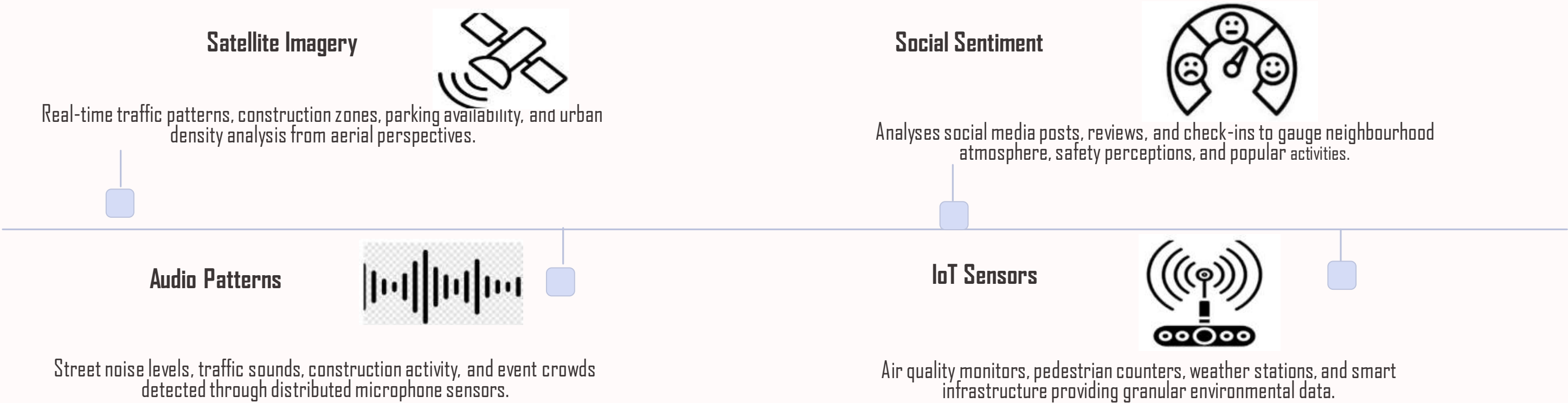


Delivers guidance through voice instructions, visual AR overlays, and haptic notifications on connected devices.

Representative Sample

3.Inspiration: Urban Pulse Navigator

Traditional navigation applications optimise for a single factor—usually fastest route by time or distance. However, urban experiences encompass far more dimensions: air quality, crowd density, cultural events, noise levels, and the subjective "energy" of different neighbourhoods.



Intelligent Urban Experience

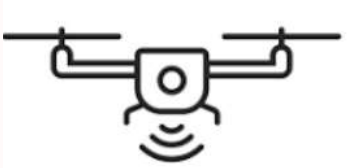
UrbanPulse doesn't just navigate around traffic—it helps you discover the optimal route for your specific needs at that moment. Seeking a peaceful walk? It routes you through quieter streets with better air quality. Meeting friends? It suggests areas with vibrant energy and nearby activities.

The system learns your preferences over time, understanding that you prefer tree-lined streets in the morning but bustling areas in the evening, or that you prioritise air quality when cycling but entertainment options when walking.

Representative Sample

4.Inspiration: PlantWhisperer Pro

Traditional farming relies heavily on visible symptoms of crop stress or disease—by which point significant damage has often occurred. PlantWhisperer Pro detects problems at their earliest stages by monitoring crops across multiple data dimensions simultaneously.



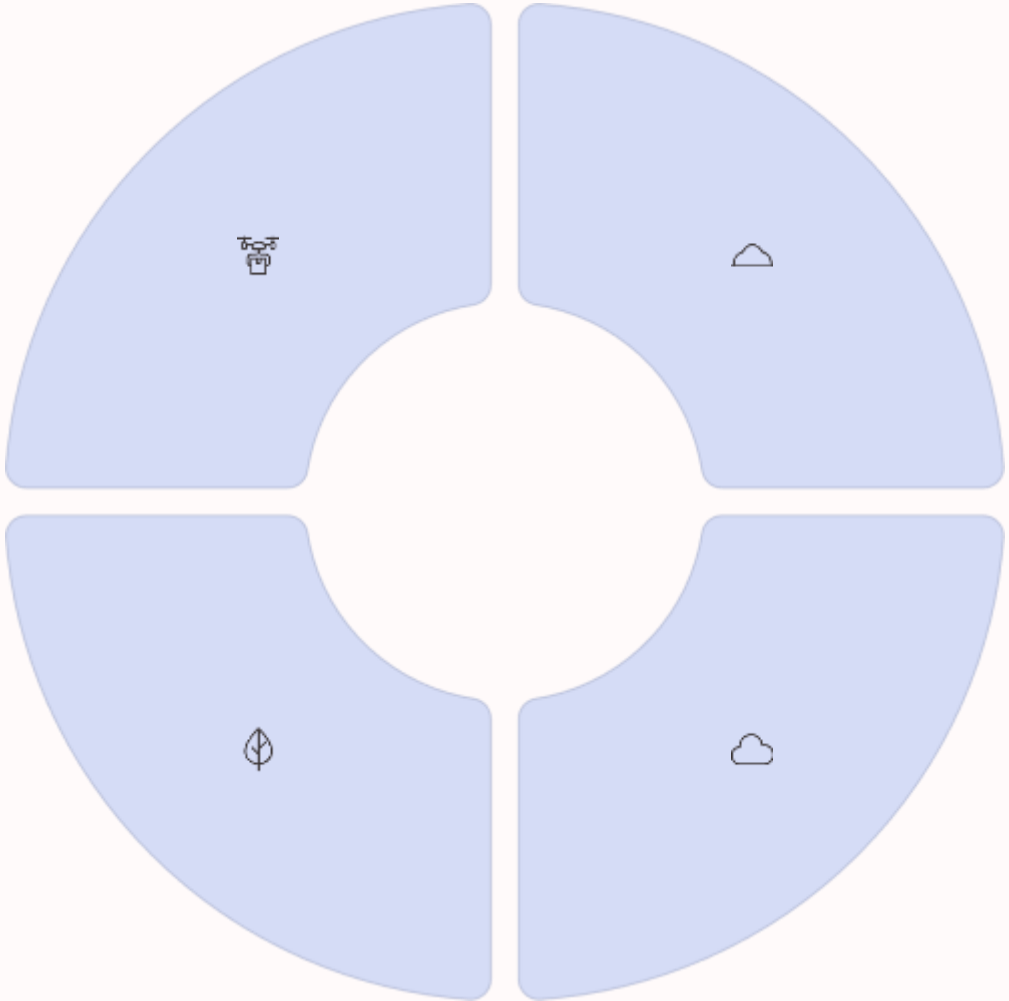
Drone Imagery

Multispectral and thermal imaging captures subtle changes in crop health invisible to human eyes, including stress patterns and disease signatures.



Plant Acoustics

Ultrasonic sensors detect stress sounds emitted by plants experiencing water shortage or pest damage—often days before visible symptoms appear.





Soil Sensors

Continuous monitoring of moisture levels, nutrient content, pH balance, and microbial activity provides root-level insights into crop conditions.



Weather Data

Hyperlocal weather forecasting, historical climate patterns, and microclimate variations help predict crop responses and optimise interventions.

Proactive Agricultural Intelligence

By synthesising these diverse data streams, PlantWhisperer Pro enables truly proactive agriculture. Farmers receive alerts about developing issues with specific location data and actionable recommendations.

The system optimises irrigation schedules based on actual plant needs rather than fixed timers, predicts disease outbreaks before visible infection, and detects pest infestations through subtle acoustic signatures.

Hackathon Structure

The hackathon unfolds across two distinct phases, each designed to support your journey from initial concept to functional prototype. This structured approach ensures adequate time for ideation, development, and refinement.



Phase 1
Problem Exploration
Duration: 5 Days

Phase 1: Problem Exploration (5 Days)

Theme Announcement

Official hackathon launch with theme reveal and comprehensive orientation session covering expectations, resources, and guidelines.

Develop Problem Statement

Research, ideate, and refine your problem statement. Consult with mentors, explore technical feasibility, and crystallise your approach.

Submit Presentation

Prepare and submit a report and video demonstration articulating your problem, articulating your problem, proposed solution, multimodal approach, and expected impact.

Team Selection

Faculty and industry experts evaluate submissions. Approximately 20 teams advance to the development phase based on innovation, feasibility, and impact potential.



Phase2
Development
Duration: 9 Days

Phase 2: Development (9 Days)

Build Prototype

Intensive development period where shortlisted teams transform concepts into working prototypes. Access technical resources and development environments.

Midway Checkpoint

Progress review with assigned mentors. Receive feedback, troubleshoot challenges, and refine your approach based on expert guidance.

Final Submission

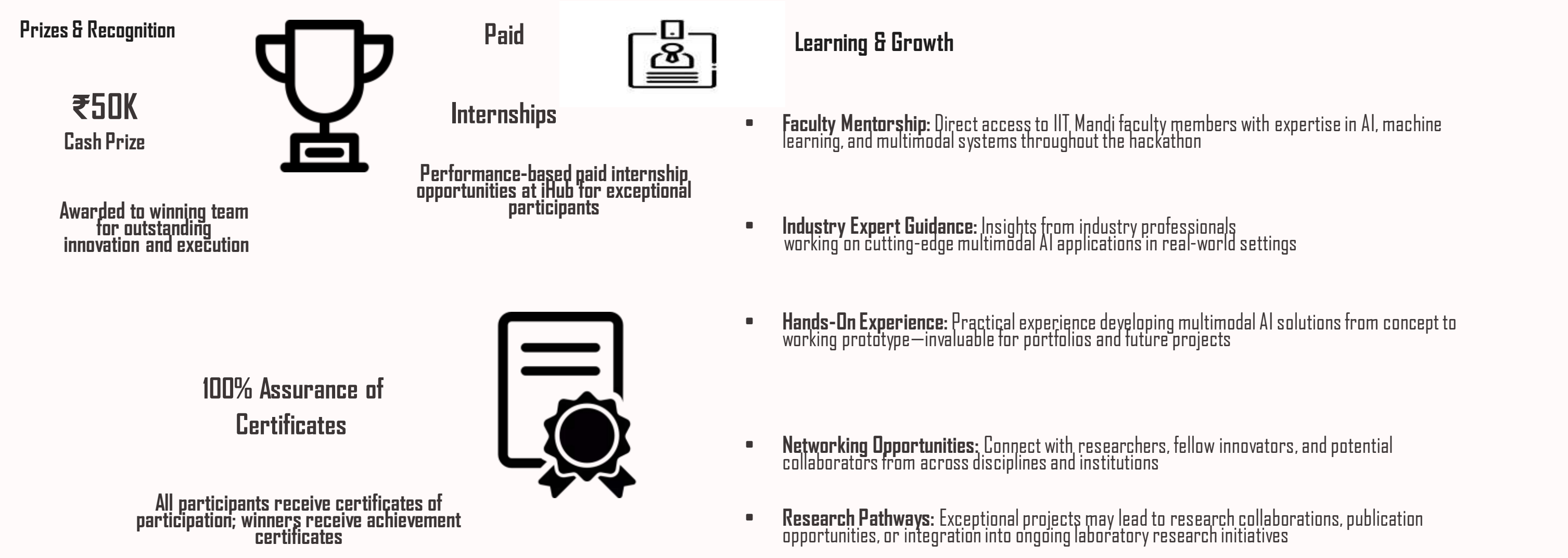
Submit your completed prototype with documentation, demonstration video, and technical specifications for comprehensive evaluation.

Pitch & Award Ceremony

Present your solution to judges through live demonstrations and structured pitches. Winners announced and prizes awarded in celebration event.

What's In It For You?

Participation offers substantial benefits extending well beyond the hackathon itself—from immediate prizes to long-term career opportunities and professional development within the AI research community.



Beyond tangible rewards, participants gain deep insight into multimodal AI development methodologies, experience collaborative innovation under time constraints, and build relationships with the IIT Mandi research community that can shape future academic and professional trajectories.



What Makes a Good Multimodal Solution?

01

Integration over Addition

Modalities should work together, not just exist side-by-side

02

Context Awareness

Understanding situations through multiple data streams

03

Real-world Impact

Solving actual problems people face

04

Indian Context

Solutions relevant to our challenges and opportunities

05

Feasibility

Can be prototyped with available tools and resources

JOIN HACKATHON



SCAN QR for Registration

Registration Link: <https://unstop.com/hackathons/multi-modality-iit-mandi-1573803>



SCAN QR for WhatsApp Community

Join WhatsApp Group for
Updates: <https://chat.whatsapp.com/IQLSZ6VVNio465d9h3C3Dj>



IIT Mandi iHub & HCl Foundation
Technology in Harmony with Human Needs



MULTIMODAL LAB
A symphony of senses in the digital realm



IIT Mandi iHub and HCl Foundation



iHub IIT Mandi – Multimodal Lab

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